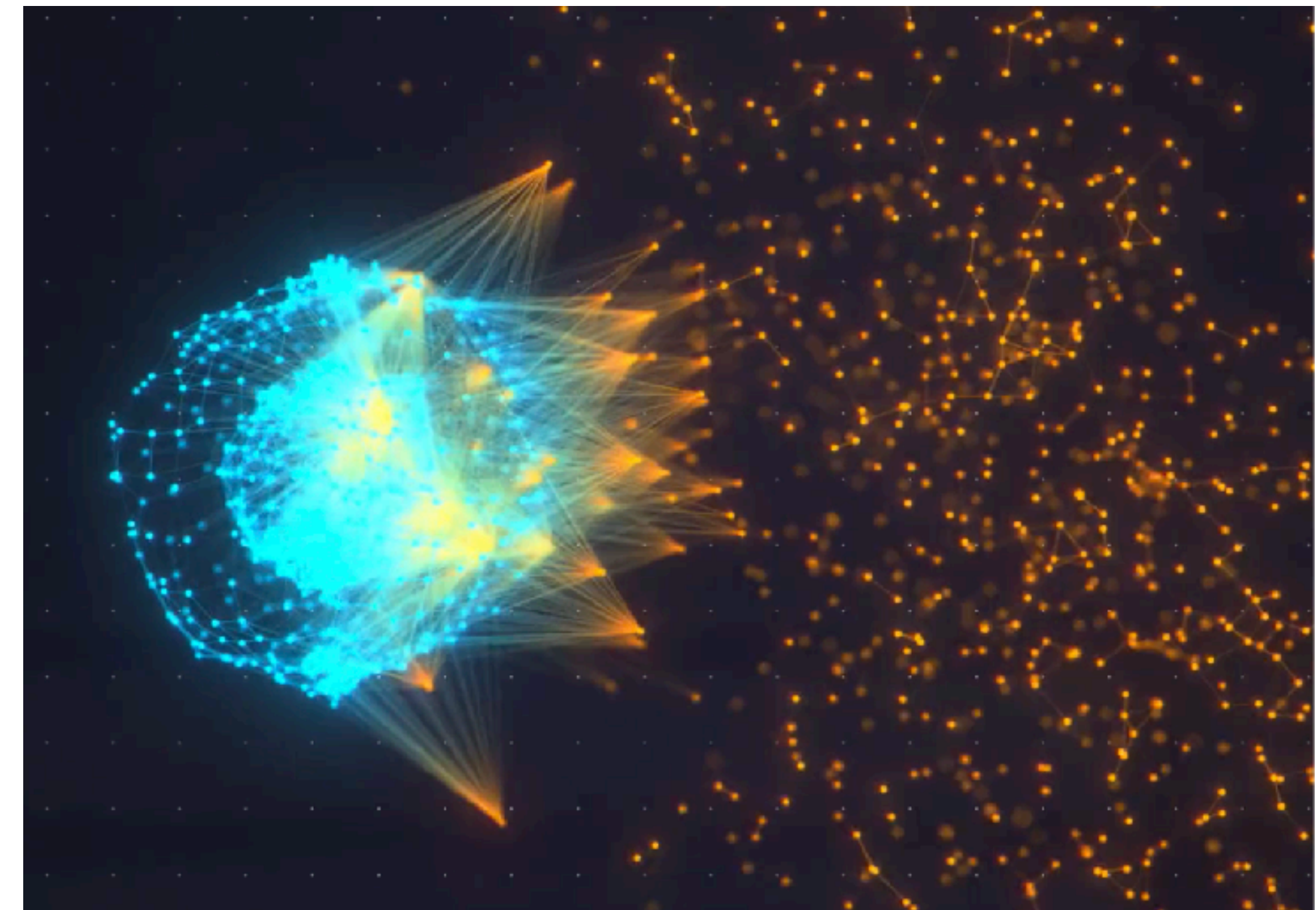


Lecture 1: Course introduction & cold plunge

Instructors: Anastassia Vybornova & Michael Szell

Aug 26, 2025

Start at 14:15



Today we will prepare the ground for an intensive course!

Part 1

1. Meet the team
behind this course



2. Course contents:
schedule, organisation



3. Python Crash Course:
why & how?



Part 2



1. Meet the team

Meet the team

2 teachers

Anastassia
(she/her)



Michael
(he/him)



+ 5 teaching assistants (TAs)

Alexandru
(he/him)



Madeeha
(she/her)



Nielson



Tetiana
(she/her)



Vincenzo
(he/him)



Meet the StudyLab assistants

Katya



Mirka
(she/her)



Mykyta
(he/him)



In case of issues, contact:

Michael



misz@itu.dk

Anastassia



anvy@itu.dk

Anonymous feedback form

Student Affairs and Programmes (SAP) located in 2A30

<https://itustudent.itu.dk/study-administration/contact-study-administration>

Meet Anastassia

Postdoc in Computational Social Science @ KU

Visiting researcher & external lecturer @ ITU

Visiting researcher @ Complexity Science Hub Vienna

**Chair of the Young Academy Panel,
Danish Data Science Academy (DDSA)**



Meet Anastassia



Postdoc in Computational Social Science @ KU

Visiting researcher & external lecturer @ ITU

Visiting researcher @ Complexity Science Hub Vienna

Chair of the Young Academy Panel,
Danish Data Science Academy (DDSA)

my coding journey html → C++ → MATLAB → R → Python → ...

Meet Anastassia

Artwork by Karo Berghuber



Aiello et al. (PNAS 2025):
Urban Highways Are
Barriers for Social Ties

sustainable transportation

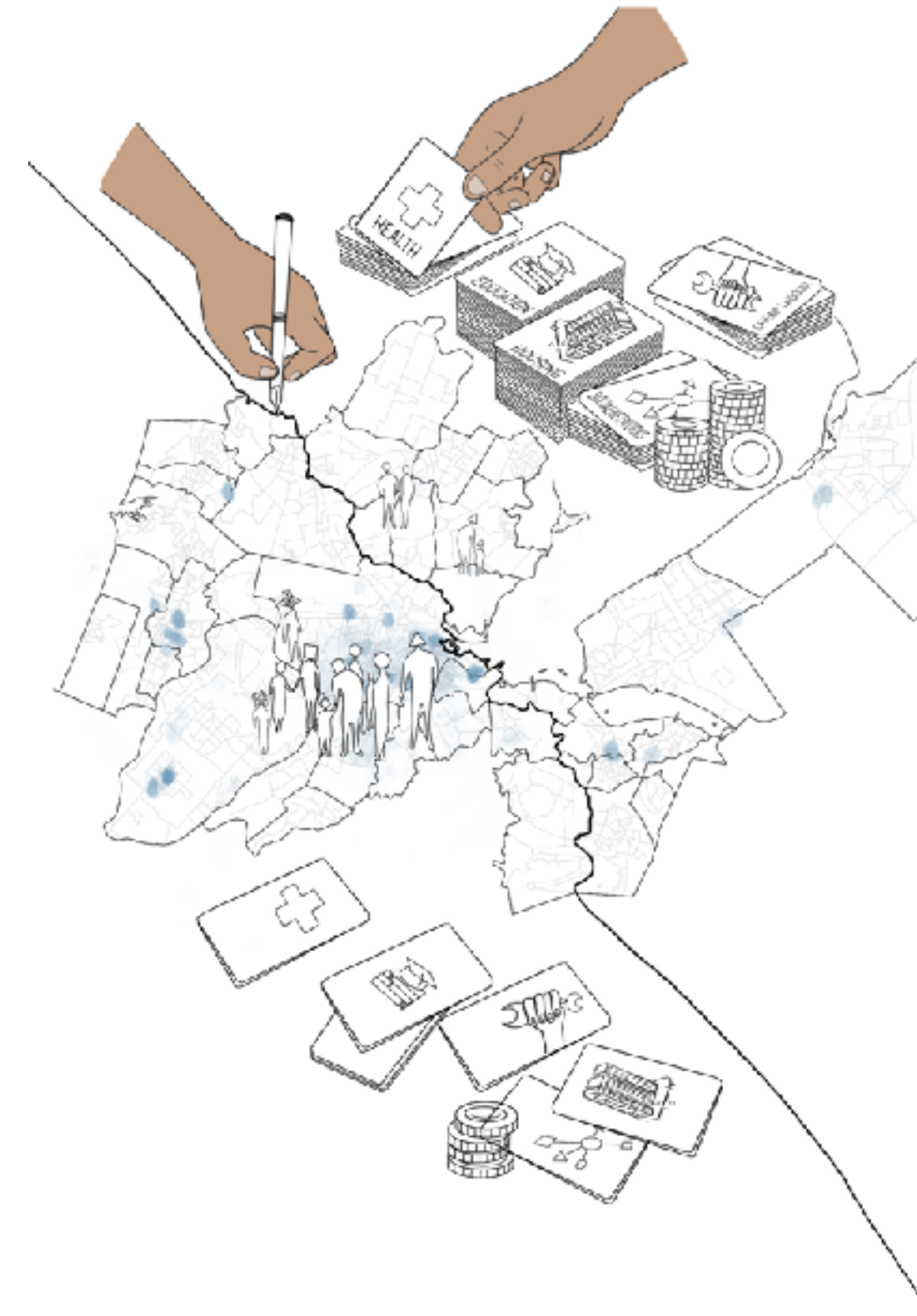
urban inequalities

geospatial data science

intersection of social and
spatial networks

algorithmic bias in urban
policy-making

Artwork by Namrata Narenda



Vybornova & Verma (arXiv 2025):
You Don't Have to Live Next to Me:
Towards A Demobilisation of
Individualistic Bias in
Computational Approaches to
Urban Segregation



Meet each other!

menti.com

7151 3902

2. Overview of course contents

First you will learn Python; then comes everything else



Lecture	Instructor	CalW	CourseW	Time	Date	Day	Place	Topic	Event
1	A	35	1	14:00-16:00	Aug 26	Tue	Aud 2	Python Crash Course	
2	A	35	1	12:00-14:00	Aug 28	Thu	Aud1		
3	A	36	2	14:00-16:00	Sep 2	Tue	Aud 2		
4	A	36	2	12:00-14:00	Sep 4	Thu	Aud1		
5	A	37	3	14:00-16:00	Sep 9	Tue	Aud 2		
6	A	37	3	12:00-14:00	Sep 11	Thu	Aud1		A1 submission
7	A	38	4	14:00-16:00	Sep 16	Tue	Aud 2		
8	A	38	4	12:00-14:00	Sep 18	Thu	Aud1		A2 submission
9	A	39	5	14:00-16:00	Sep 23	Tue	Aud 2		
10	A	39	5	12:00-14:00	Sep 25	Thu	Aud1		A3 submission
11	A	40	6	14:00-16:00	Sep 30	Tue	Aud 2		CODING TEST
12	M	40	6	12:00-14:00	Oct 2	Thu	Aud 1	Exploratory Data Analysis	
13	M	41	7	14:00-16:00	Oct 7	Tue	Aud 2		
14	M	41	7	12:00-14:00	Oct 9	Thu	Aud 1		A4 submission
Holiday week		42		Holiday week					
15	M	43	8	14:00-16:00	Oct 21	Tue	Aud 2	Exploratory Data Analysis	
16	M	43	8	12:00-14:00	Oct 23	Thu	Aud 1	Program Design	CODING RE-TEST
17	M	44	9	14:00-16:00	Oct 28	Tue	Aud 2		
18	M	44	9	12:00-14:00	Oct 30	Thu	Aud 1		A5 submission
19	M	45	10	14:00-16:00	Nov 4	Tue	Aud 2		
20	M	45	10	12:00-14:00	Nov 6	Thu	Aud 1	Networks & Skewed Data	A6 submission
21	M	46	11	14:00-16:00	Nov 11	Tue	Aud 2		
22	M	46	11	12:00-14:00	Nov 13	Thu	Aud 1		
23	M	47	12	14:00-16:00	Nov 18	Tue	Aud 2		
24	M	47	12	12:00-14:00	Nov 20	Thu	Aud 1	Data Science in Practice	A7 submission
25	M	48	13	14:00-16:00	Nov 25	Tue	Aud 2		
26	M	48	13	12:00-14:00	Nov 27	Thu	Aud 1		
27	M	49	14	No lecture	Dec 2	Tue	-	Mock Exam	
28	M	49	14	12:00-14:00	Dec 4	Thu	Aud 1		
		50		09:00-13:00	Dec 11	Wed	TBD	Final Exam	EXAM



3. Course organisation

Our weekly schedule - overview

	Monday	Tuesday	Wednesday	Thursday	Friday		
8							
9							
10							
11							
12					Lecture Aud 1	Live coding Aud 2	
13							
14				Lecture Aud 2		Exercises 2A52 & 2A54	StudyLab 3A50
15							
16				Exercises 2A52 & 2A54		StudyLab 3A50	
17							

Our interface to this course is LearnIT

The screenshot shows the LearnIT interface. At the top, there's a header with 'ITU LEARNIT' and navigation links: 'Dashboard', 'My courses', and 'All courses'. On the right of the header are icons for search, notifications, and a user profile, along with an 'Edit mode' toggle. The left sidebar contains a menu with 'About this course' (expanded) showing 'News', 'Q&A Forum', 'Ordinary exam assignment', 'Re-exam assignment', and 'Examination Syllabus'. Below this are 'Python Crash Course: Res...', 'Lecture 01 (Aug 26)', 'Lecture 02 (Aug 28)', 'Lecture 03 (Sep 02)', 'Lecture 04 (Sep 04)', 'Lecture 05 (Sep 09)', and 'Lecture 06 (Sep 11)'. The main content area has a banner for 'Introduction to Data Science and Programming (Autumn 2025)' with a video player below it showing 'Data Scientist researching:'. Above the video player is a navigation bar with 'Course', 'Settings', 'Participants', 'Grades', 'Reports', and 'More'. Below the video player, there's a section titled 'About this course' with a 'Collapse all' link. The right sidebar, titled 'People', lists the 'COURSE MANAGER' (Michael Szell), 'TEACHER' (Anastassia Vybornova), and 'TEACHING ASSISTANT' (Alexandru Jizdan, Madeeha Masood, Vincenzo Emanuele Piras, Tetiana Tretiak, and Nielson Völk).

Set your profile picture here!

<https://learnit.itu.dk/user/edit.php>

<https://learnit.itu.dk/course/view.php?id=3024641>

Our interface to this course is LearnIT

The screenshot displays the ITU LearnIT interface. The top navigation bar includes 'ITU LEARNIT' and links for 'Dashboard', 'My courses', and 'All courses'. The sidebar on the left contains a list of course-related items, with 'Lecture 01 (Aug 26)' highlighted. The main content area shows the 'Course' tab selected, with a list of topics covered in Lecture 01. A pink oval highlights the 'Exercise 01' link in the sidebar and the 'Exercise 01' link in the main content area.

ITU LEARNIT Dashboard My courses All courses

Course Settings Participants Grades Reports More ▾

▼ About this course

News

Q&A Forum

Ordinary exam assignment

Re-exam assignment

Examination Syllabus

▼ Python Crash Course: Res...

▼ Lecture 01 (Aug 26)

Exercise 01

▼ Lecture 01 (Aug 26)

Topics covered:

- Course intro & organization
- Python basics: operators, objects & their types, variables & value assignment; using Jupyter notebook

Exercise 01

▼ Lecture 02 (Aug 28)

Here we will distribute
all learning material

Python Crash Course: learning materials on GitHub first

anastassiavybornova / learn-python-2025

Type `/` to search

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

learn-python-2025 Public

Pin

Watch 0

Fork 0

Starred 3

main 1 Branch 0 Tags

Go to file

Add file

<> Code

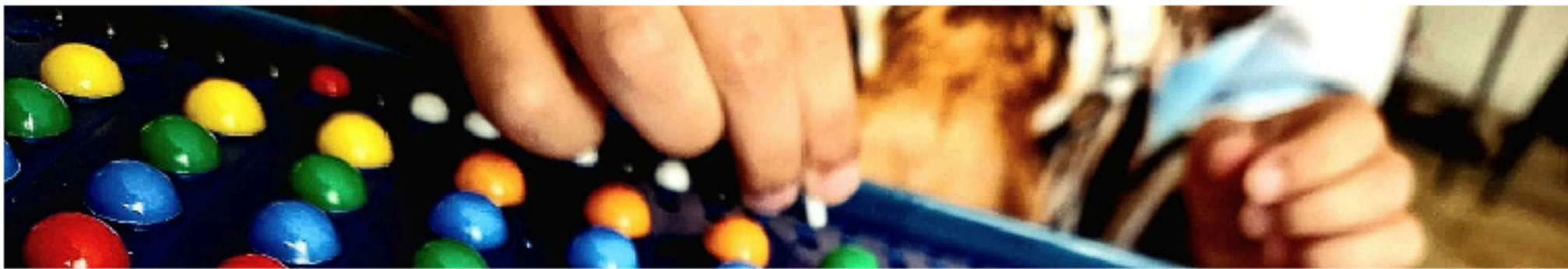
anastassiavybornova update readme + mastermind img 6950f42 · 20 hours ago 7 Commits

books	add books	yesterday
images	update readme + mastermind img	20 hours ago
unit01	add unit01 exercises	yesterday
.gitignore	update gitignore	yesterday
LICENSE	Initial commit	yesterday
README.md	update readme + mastermind img	20 hours ago

README

GPL-3.0 license

learn-python-2025



About

Python Crash Course materials for IDSP @ ITU, fall 2025

[learnit.itu.dk/course/view.php?id=30...](#)

Readme

GPL-3.0 license

Activity

3 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages

Jupyter Notebook 100.0%

Will be linked on learnIT

<https://github.com/anastassiavybornova/learn-python-2025>

We will take plenty of breaks during class

Tuesday

Lecture

14:15-16:00

with a 10min break

Exercises

16:15-18:00

with a 10min break

Thursday

Lecture

12:15-14:00

with a 10min break

Exercises

14:15-16:00

with a 10min break

We will take plenty of breaks during class

Tuesday

Lecture
14:15-16:00
with a 10min break

Exercises
16:15-18:00
with a 10min break

Thursday

Lecture
12:15-14:00
with a 10min break

Exercises
14:15-16:00
with a 10min break



We are MANY people.
Please be on time!

We will have some **mandatory** activities

- 7 Assignments** Similar to exercises
 - Distributed through learnIT on Thursdays (first one next week, Sep 4)
 - Due 1 week later, Thursday 23:59
 - For each assignment one second attempt is granted.
 - We allow for a max of 2 assignments to be failed/not submitted.

- 1 Coding Test** End of September 2025
 - Second chance end of October 2025

- 1 Final Written Exam** Dec 11, 2025

You need to pass both the assignments & the coding test to be admitted to the written exam!

3. Why Programming? Why Python?

How to enjoy the Python Crash Course?

Data Scientists need 4 skills

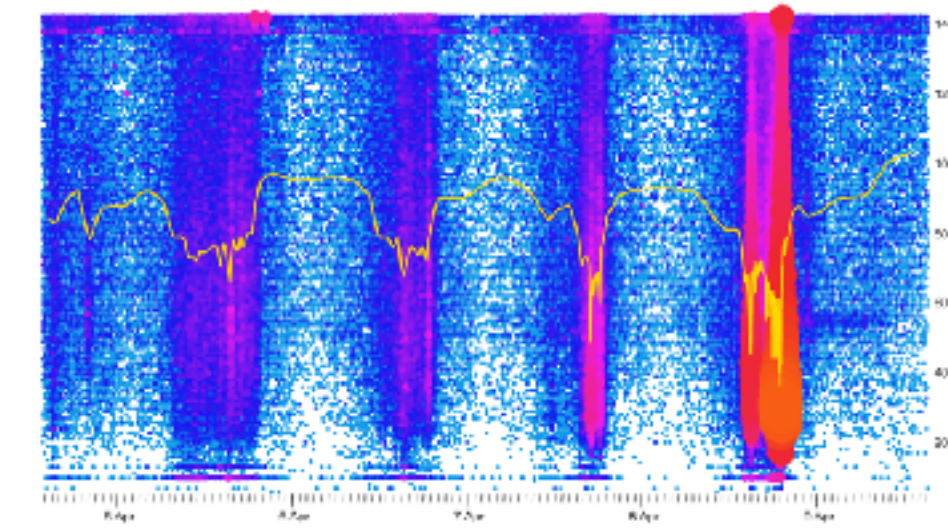
1) Programming & Computer Science
to handle large data sets efficiently



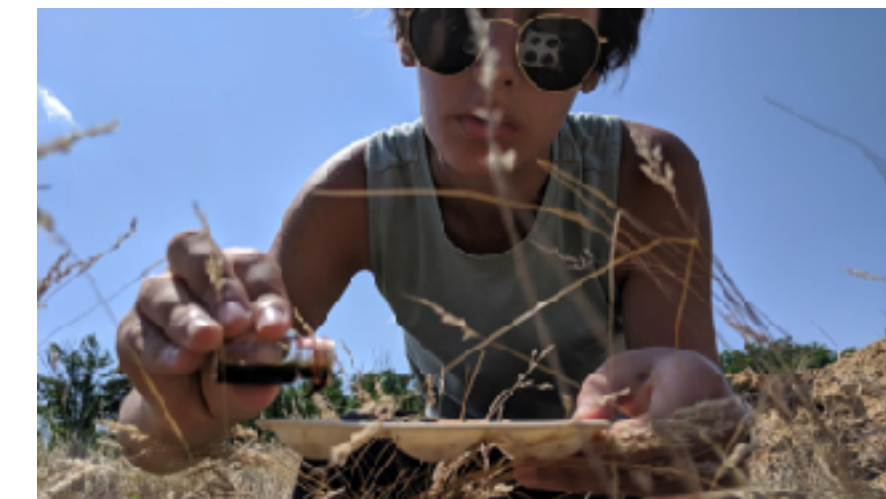
2) Mathematics & Statistics
to correctly ask questions and analyze data

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

3) Communication & Visualization
to exchange insights, especially with somebody less conversant in data



4) Domain knowledge
to ask the right questions



Data Scientists need 4 skills

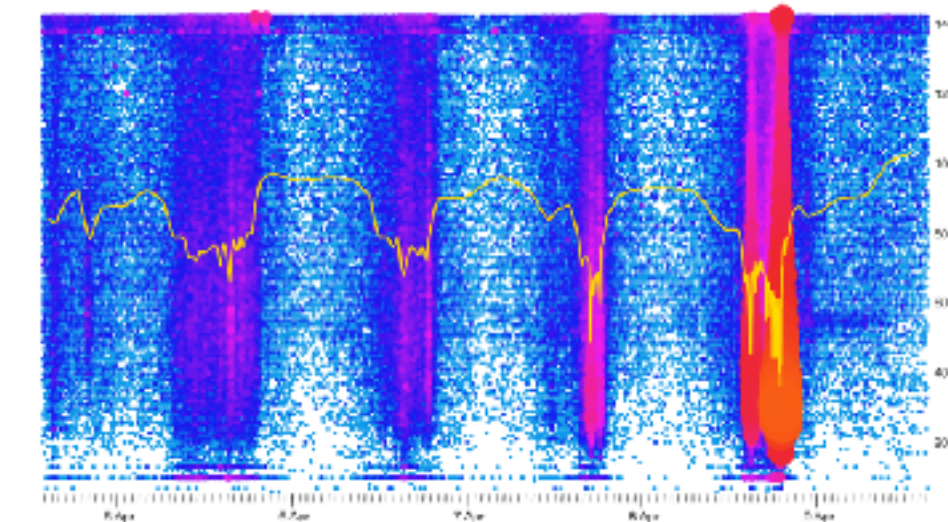
1) Programming & Computer Science
to handle large data sets efficiently



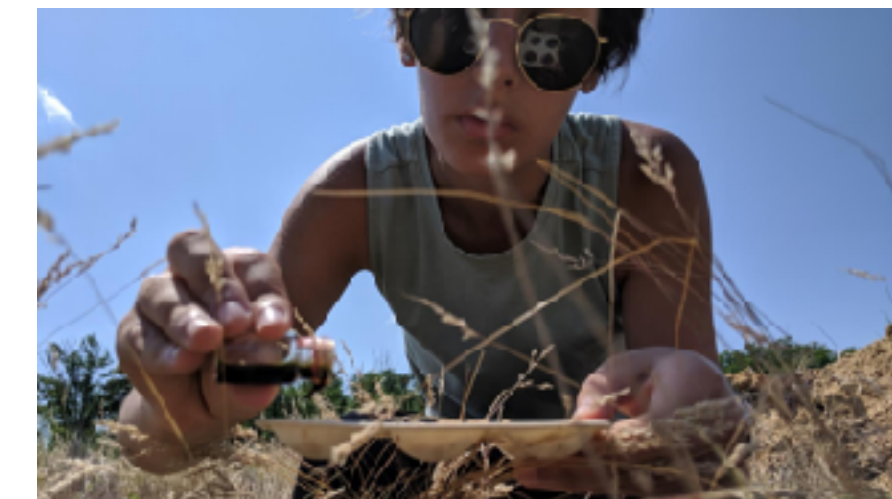
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3) Communication & Visualization
to exchange insights, especially with somebody less conversant in data



4) Domain knowledge
to ask the right questions



Starting your programming journey with Python because:

easy



Starting your programming journey with Python because:

~~easy~~



Starting your programming journey with Python because:

~~easy~~ you can start coding right away



Starting your programming journey with Python because:

~~easy~~ you can start coding right away

if you've never coded before,
many recommend to start with Python



Starting your programming journey with Python because:

~~easy~~ you can start coding right away

if you've never coded before,
many recommend to start with Python

once you have a “coding mindset”, it’s much
easier to learn a new programming language



Starting your programming journey with Python because:

~~easy~~ you can start coding right away

if you've never coded before,
many recommend to start with Python

once you have a “coding mindset”, it’s much
easier to learn a new programming language

widely used, particularly for Data Science



Be prepared to start coding at the next lecture on Thursday!

!!! Go to the **exercises** today (Exercise01: “bootIT”)

1. Check out the installation instructions (bootIT)
2. Verify that it worked by running a notebook (testIT)

!!! Next lecture, we will already be using jupyter notebook in class

- **be prepared!**

Be prepared to start coding at the next lecture on Thursday!

!!! Go to the **exercises** today (Exercise01: “bootIT”)

1. Check out the installation instructions (bootIT)
2. Verify that it worked by running a notebook (testIT)

!!! Next lecture, we will already be using jupyter notebook in class

- **be prepared!**

(If you're not - you can still get help in the Study Labs (first one is on Aug 28))

Where do I get support?

Lectures

Exercises

Study Lab

Books

The Q&A forum in learnIT

Your peers

Live coding sessions

Last resort: the internet! Looking up error messages; checking online forums (e.g. stackoverflow.com)



Exercises & supplementary activities

Exercises: Following each lecture: Every Tue 16:00-18:00 and Thu 14:00-16:00 (group information to follow), Rooms 2A52 & 2A54

Live Coding: Every Fri 12:30-14:00, Room Aud 2

Study Lab: Every Thu 16:00-18:00 and Fri 14:00-16:00, Room 3A50



Feedback / Contact

Anonymous feedback form: <https://forms.gle/J83AMKiAwChm9Qm17>, or ITU's [heretohelp](#) page



News



Q&A Forum

There are some great books to help you learn how to code!

✓ Python Crash Course: Resources

3 books we recommend for learning Python, all freely available as PDFs:

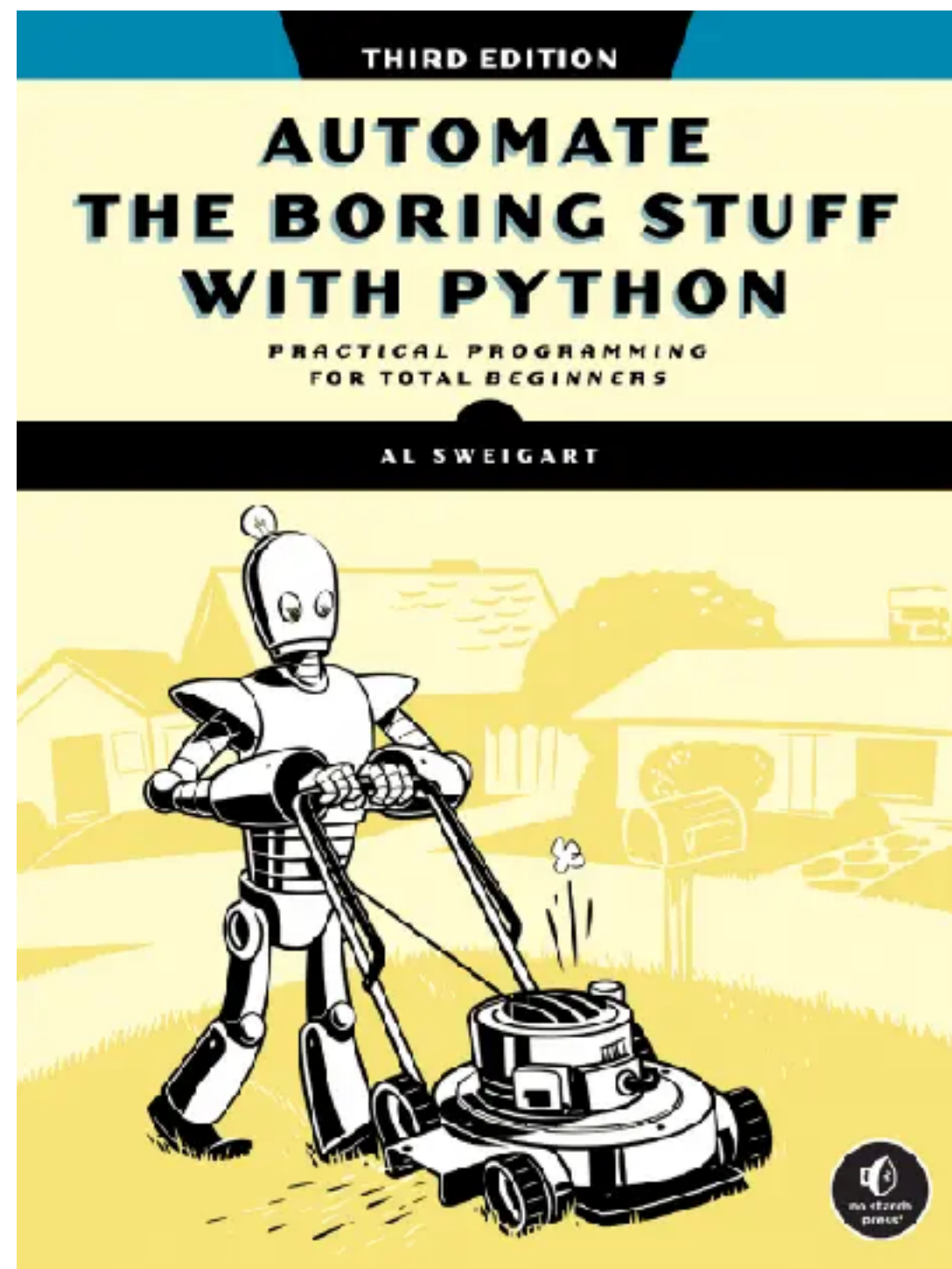
- **AS:** Al Sweigart's [Automate the Boring Stuff](#)
- **EM:** Eric Matthes' [Python Crash Course](#) (3rd edition)
- **CM:** Connor Milliken's "[Python Projects for Beginners: A Ten-Week Bootcamp Approach to Python Programming](#)"
(log in with your ITU credentials to access the PDF; or get the PDF [here](#).)

We use the book abbreviations (AS, EM, CM) to point you to chapters of each book, corresponding to some lecture/exercise content.

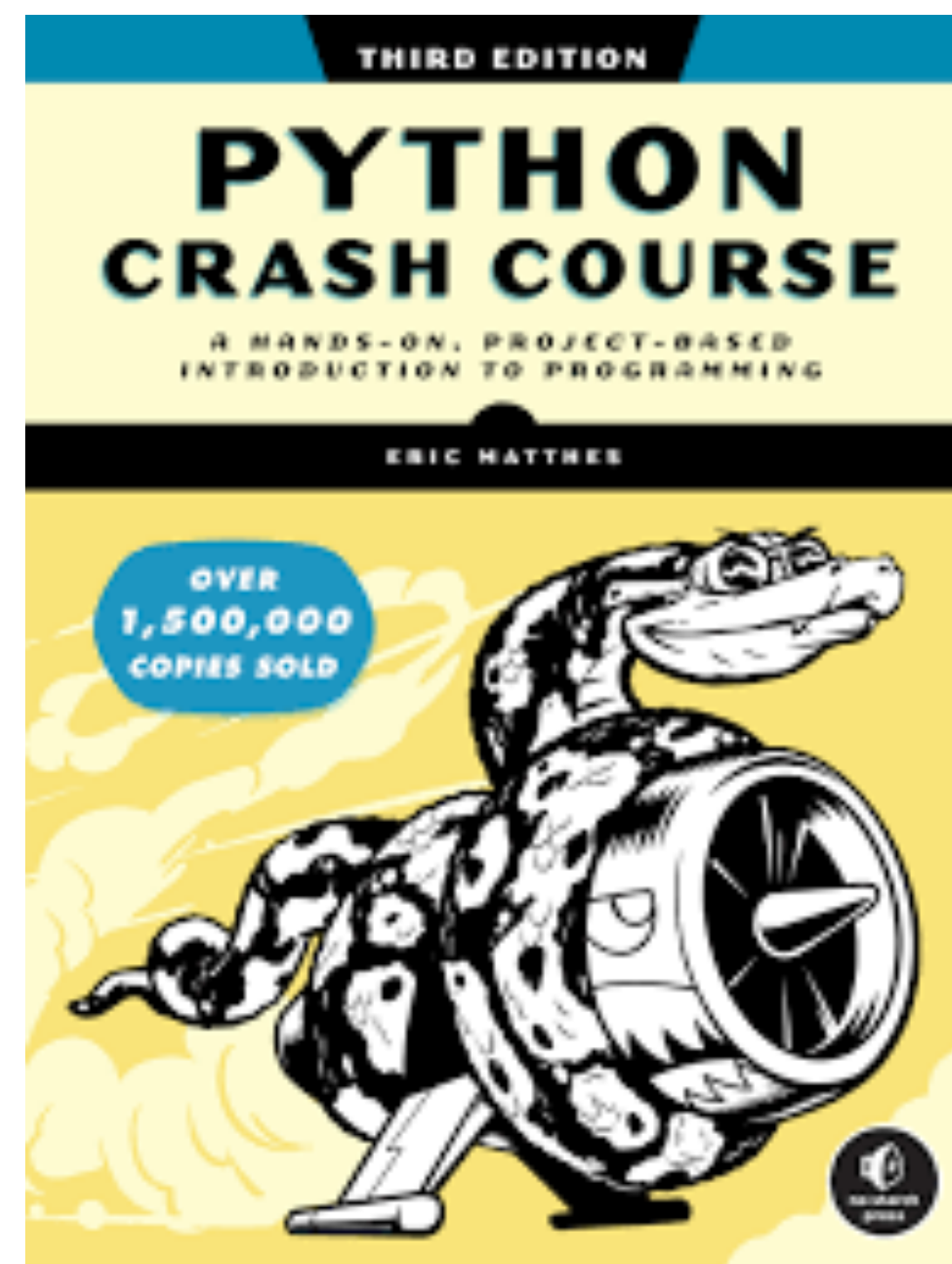
Recommended online Python course with a computer science focus: [MIT 6.0001: Introduction to Computer Science and Programming in Python](#) (Ana Bell & Eric Grimson & John Guttag)

Resources will be updated throughout the course.

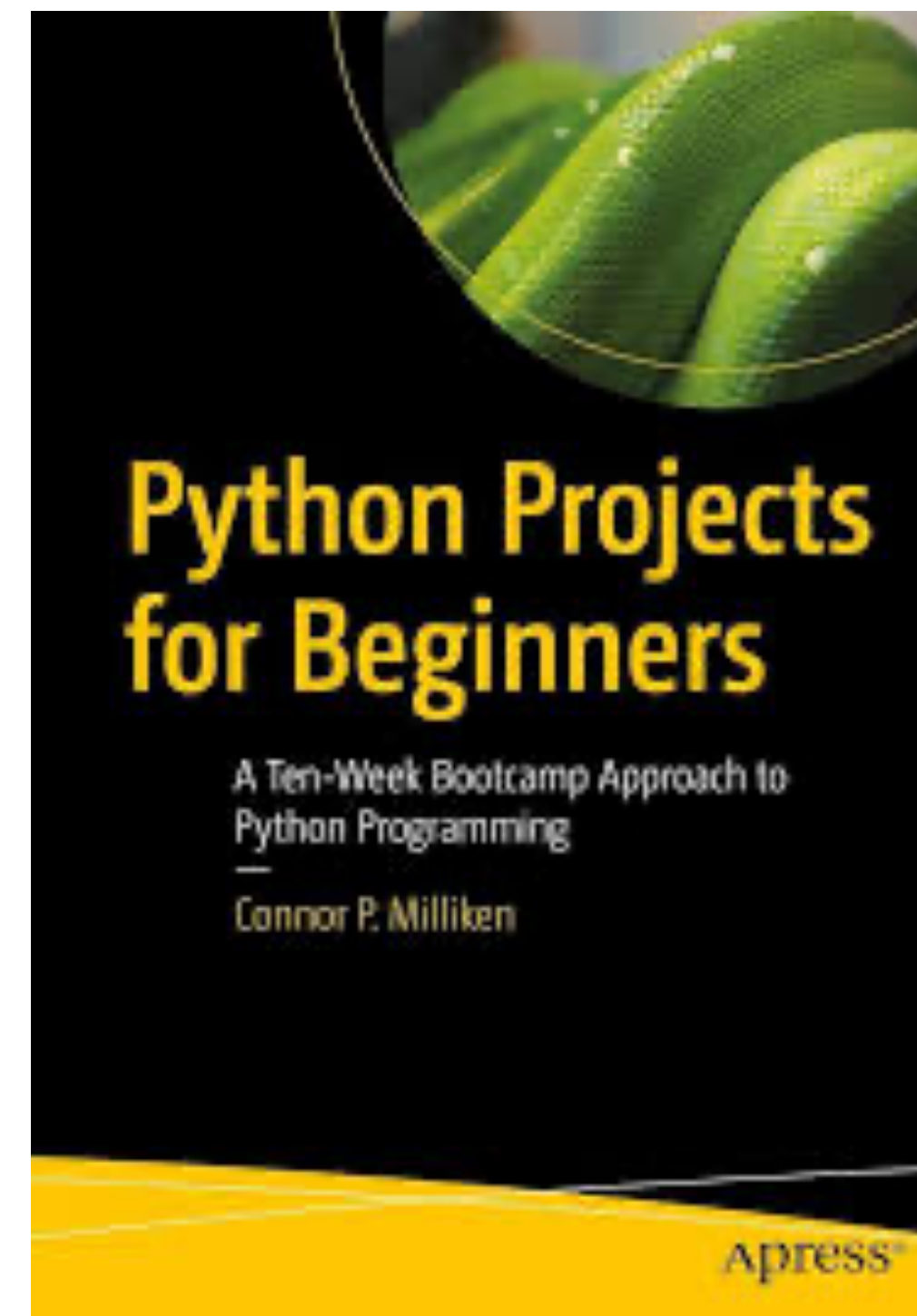
There are some great books to help you learn how to code...



AS



EM



CM

...and some great online courses!

Introduction To Computer Science And Programming In Python

Syllabus

Instructor Insights

Readings

Lecture Videos

Lecture Slides and Code

In-Class Questions and
Video Solutions

Assignments

Lecture Videos

Lecture 1: What is Computation?

Description: In this lecture, Dr. Bell introduces the theory of computation and explains some aspects of computational thinking. Programming languages are discussed, with an emphasis on basic Python syntax and data structures.

Instructor: Dr. Ana Bell



Course Info



INSTRUCTORS

[Dr. Ana Bell](#)
[Prof. Eric Grimson](#)
[Prof. John Guttag](#)

DEPARTMENTS

[Electrical Engineering and Computer
Science](#)

AS TAUGHT IN

Fall 2016

LEVEL

[Undergraduate](#)

TOPICS

▼ [Engineering](#)

▼ [Computer Science](#)

[Algorithms and Data Structures](#)

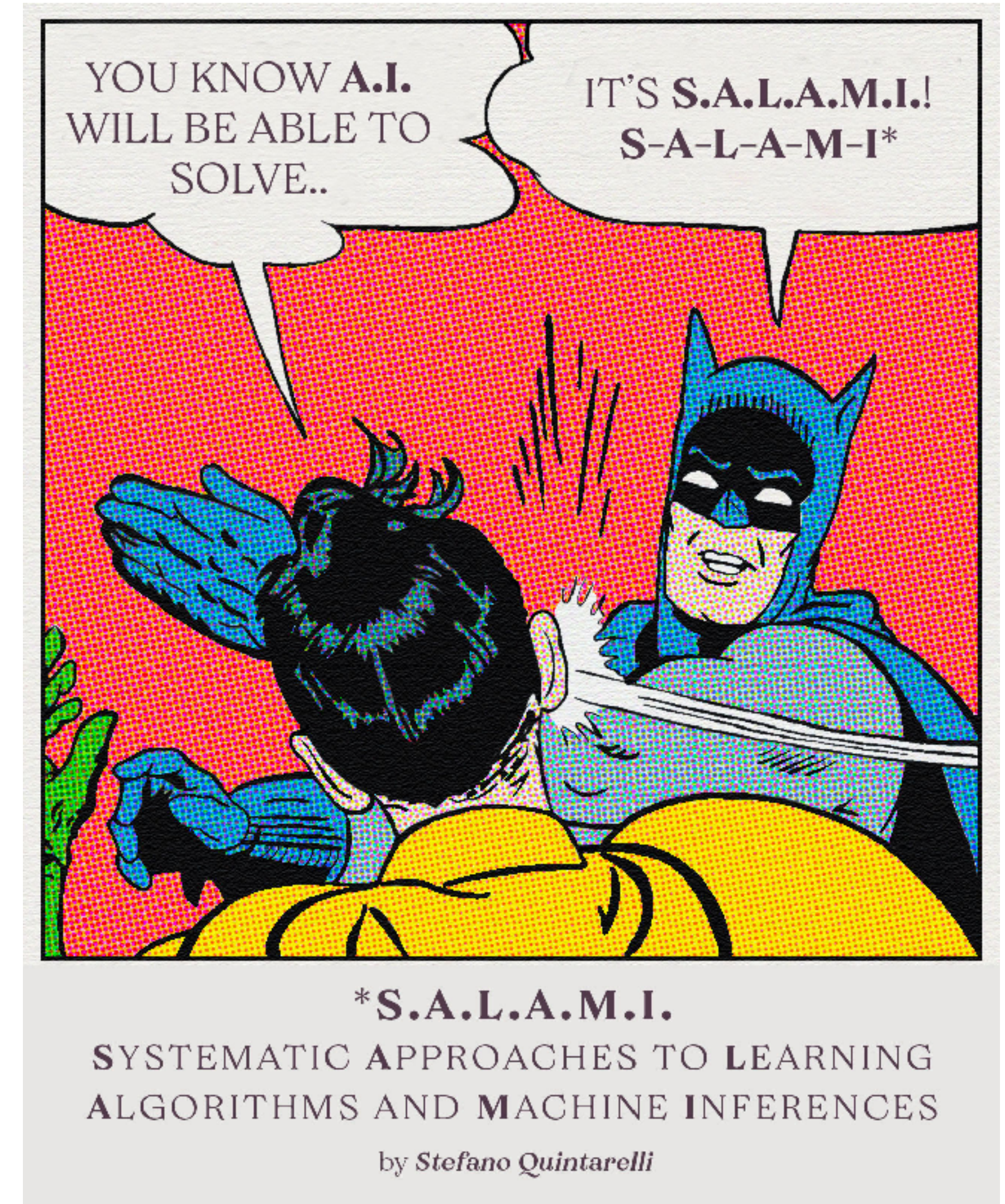
[Programming Languages](#)

LEARNING RESOURCE TYPES

[Problem Sets](#)

[Lecture Notes](#)

What happens if I use SALAMI (GenAI) to pass this class?



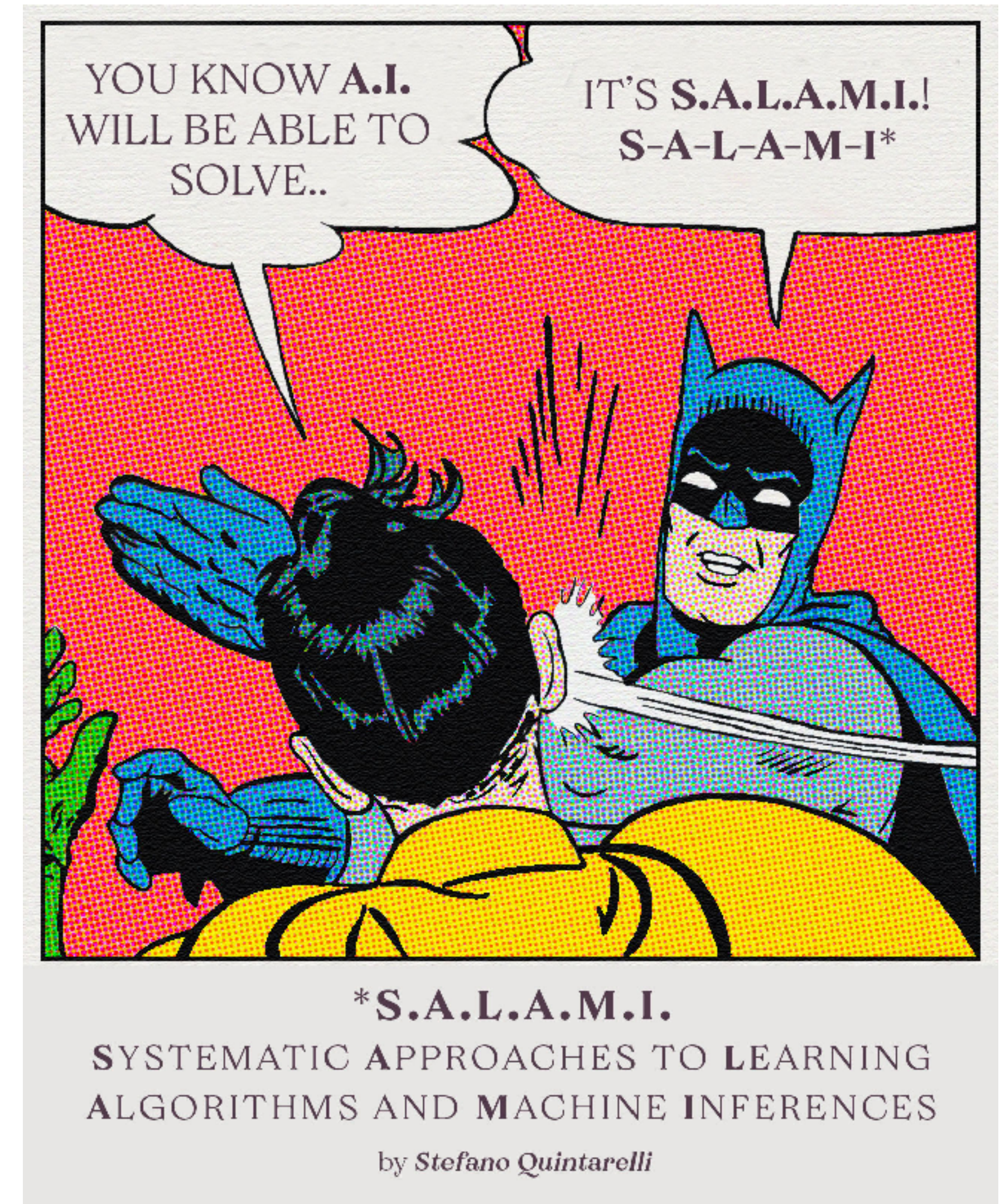
<https://blog.quintarelli.it/2019/11/lets-forget-the-term-ai-lets-call-them-systematic-approaches-to-learning-algorithms-and-machine-inferences-salami/>

What happens if I use SALAMI (GenAI) to pass this class?

Use of generative AI during the **exam**: No use of generative artificial intelligence is allowed at the exam under any circumstances

Use of generative AI for the **assignments**: Use of generative AI for assignments is not forbidden, but needs to be declared in the submitted collaborators.txt file. Last year's use of genAI by some students for assignments did not show benefits, so we discourage its use.

Use of generative AI during the **lectures/exercises**: ...Talk to us instead!



<https://blog.quintarelli.it/2019/11/lets-forget-the-term-ai-lets-call-them-systematic-approaches-to-learning-algorithms-and-machine-inferences-salami/>

What happens if I use SALAMI (GenAI) to code, e.g. at work?

It is well-known that LLMs sometimes produce verbatim or near-verbatim copies of their training data, most of which cannot be used without attribution (and may have more onerous license requirements). It is also well-known that they don't "understand" semantics: they never make changes for the right reason.

Imagine you have a supremely-untrustworthy colleague who is repeatedly caught plagiarizing and lying, is never able to explain their work, but has not been fired due to superficial sycophancy. You ask them to write a function and you know they have samples of incompatibly-licensed code for similar functions on their screen while they are working (but you don't know affirmatively which samples), you know they lie pathologically and will try to obfuscate, and you know they don't have the skills and understanding to write the function independently. Yet they submit code that you determine is functional.

Can you sign off on it on their behalf, with your reputation and legal liability on the line, and the health of your community at stake?



Jed Brown

@jedbrown@hachyderm.io

<https://hachyderm.io/@jedbrown/114928816429127795>

**BONUS: Life hacks
for this course**

BREAK UNTIL 15:15

A polar bear is swimming in the water, with its head and front paws visible above the surface. The bear's fur is white, and its eyes are closed. The water is a deep blue color.

jupyter.org/try-jupyter/lab